

AVISH PRADHAN

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Summary

B.Tech Computer Science student (2026) focused on backend development and algorithmic problem solving. Built real-time full-stack applications using Node.js, Express, and REST APIs, including secure authentication systems and optimized database queries. Solved 150+ DSA problems on LeetCode and GFG, strengthening core problem-solving skills. Actively seeking backend or SDE internship opportunities.

Skills

Programming: C++, Python, Java, SQL

Backend: Node.js, Express.js, REST APIs

Databases: PostgreSQL, SQLite

Core Concepts: Data Structures & Algorithms, Authentication (JWT, TOTP), System Design Basics

Tools: Git, GitHub, Render

Frontend: HTML, CSS, JavaScript, Bootstrap

Projects

SecureAuth — Two-Factor Authentication Web App

GitHub

Flask, PyOTP, SQLite, JWT, TOTP, HTML/CSS

- Built a secure authentication system using Flask implementing TOTP-based 2FA and JWT-based session management
- Designed stateless authentication with token-based access control and session expiry handling
- Reduced authentication latency through optimized token validation and efficient session handling
- Structured modular backend architecture to improve maintainability and scalability

RouteWise AI — AI-Powered Multi-Stop Routing System

Python (FastAPI), Scikit-learn, Genetic Algorithms, Leaflet.js

- Engineered a multi-stop route optimization system using a Genetic Algorithm with 2-opt heuristics, reducing average travel distance by 18%
- Integrated real-time traffic and weather APIs to enable dynamic routing across 1,000+ simulated route scenarios
- Developed a Scikit-learn regression model for delay prediction, improving accuracy from 74% to 91% using feature engineering
- Optimized backend performance using parallel API calls and LRU caching, achieving route computation in under 10 seconds

Study Planner — Task Optimization and Scheduling System

Python (Flask), SQLite, Chart.js

- Developed a study planning system managing 50+ academic tasks, reducing manual scheduling effort by 30%
- Applied Greedy and Knapsack-based optimization techniques to prioritize tasks and allocate time efficiently
- Built a Flask backend with SQLite handling 100+ daily queries, enabling real-time dashboard updates using Chart.js

Education

Year	Degree / Exam	Institute	Board	CGPA / Marks
2022 - 2026	B.Tech in Computer Science	Graphic Era Hill University, Dehradun	-	8.1 CGPA
2022	Class XII (CBSE)	The Aryan Public School, Meerut	CBSE	85%

Achievements & Certifications

- Solved 150+ Data Structures and Algorithms problems on LeetCode and GeeksforGeeks
- Built multiple backend-focused projects involving authentication, real-time systems, and optimization problems
- Gained hands-on experience with REST API design, database management, and system-level problem solving
- Completed foundational AWS modules covering IAM, cloud architecture basics, and deployment workflows